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Esercizio:

$$A=\begin{pmatrix}1 & 2 & 1\\ 0 & 1 & 0\\ -1 & 6 & h\end{pmatrix}$$

a) $\det A \neq 0$ $\quad h \neq 0$ $\quad \det(A) = 1\cdot\det\begin{pmatrix}1 & 0\\ -1 & h\end{pmatrix} = h+3$ → $h \neq -3$ $\quad \forall h \in \mathbb{R}-\{-3\}$

b) $h=0$, $A^{-1} = \frac{1}{\det(A)}\cdot(\text{coniugata})^T$

$$= \frac{1}{3}\cdot\begin{pmatrix}0 & 18 & -3\\ 0 & 3 & 0\\ 1 & -8 & 1\end{pmatrix} = \begin{pmatrix}0 & 6 & -1\\ 0 & 1 & 0\\ 1/3 & -8/3 & 1/3\end{pmatrix}$$

coniugata di A: $C_{11} = (-1)^{1+1}\cdot h = 0$

$C_{12} = (-1)^{1+2}\cdot 0 = 0$

$C_{13} = (-1)^{1+3}\cdot(-1) = -1$

$C_{21} = (-1)^{2+1}\cdot(2h-18) = 18$

$C_{22} = (-1)^{2+2}\cdot(h+3) = 3$

$C_{23} = (-1)^{2+3}\cdot(6+2) = -8$

$C_{31} = (-1)^{3+1}\cdot(-3) = -3$

$C_{32} = (-1)^{3+2}\cdot(0) = 0$

$C_{33} = (-1)^{3+3}\cdot(1) = 1$

$$\begin{pmatrix}0 & 0 & 1\\ 18 & 3 & -8\\ -3 & 0 & 1\end{pmatrix}$$

c) $X = A^{-1}\begin{pmatrix}1\\ 0\\ 5\end{pmatrix}$

$$X = \begin{pmatrix}0 & 6 & -1\\ 0 & 1 & 0\\ 1/3 & -8/3 & 1/3\end{pmatrix}\cdot\begin{pmatrix}1\\ 0\\ 5\end{pmatrix} = \begin{pmatrix}-5\\ 0\\ 2\end{pmatrix}$$

$\begin{cases}x=-5\\ y=0\\ z=2\end{cases}$ ✓

Es. 2

$P(1,0,2)$ $\quad \pi: x+y-2z+1=0$

$$r:\begin{cases}x = 1+t\\ y = -t\\ z = -2+t\end{cases} \quad t\in\mathbb{R} \rightarrow v=(1,-1,1)$$

a) P appartiene a π e non a r?

$1-2+1=0$ ✓

$\begin{cases}y = x+t \rightarrow t=0\\ 0 = -1\\ z = -2+t \rightarrow t=4\end{cases}$ Verificato

b) L'intersezione tra r e π

$1+t+2-t+1=0$

$t=4$ → $(5,-4,2)$

c) σ che contiene r e P

$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

$t=0 \rightarrow A(1,0,-2)$ $\quad \overline{PA} = (0,0,4)$

$t=1 \rightarrow B(2,-1,-1)$ $\quad \overline{PB} = (-1,1,3)$

$$\overline{PA}\times\overline{PB}: \begin{pmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 0 & 4\\ -1 & 1 & 3\end{pmatrix}\begin{matrix}i & j\\ 0 & 0\\ -1 & 1\end{matrix}$$

$\hat{i}(-4) + \hat{j}(-4) + \hat{k}(0)$

$a \quad b \quad 0$

→ $(1,1,0)$

$x - 1 + y = 0$

$\sigma: x+y=1$

d) Proiezione P su r $\quad t = 1/3$

$\overline{AP}\cdot v_r = t+t+t-4 = 3t-4$

$R-P = (t;-t;t-4)$

$P(1,0,2)$ $\quad v_r = (1,-1,1)$

$P(1+t;-t;-2+t)$

$(4/3, -4/3, -2/3)$



Es. 3

$V = \{(x,y,z,t) \mid x+2y-z=0\}$

$W = \langle (0,1,0,1), (0,0,1,1), (1,0,0,0) \rangle$

$X = \langle (1,0,-2,0), (1,-1,1,0) \rangle$

a) base $(X \cap V)$

dimensione $(X \cap V)$

$\lambda_1(1,0,-2,0) + \lambda_2(1,-1,1,0) = (\lambda_1+\lambda_2, -\lambda_2, -2\lambda_1+\lambda_2, 0)$ (z, t)

$(\lambda_1+\lambda_2) + 2(-\lambda_2) - (-2\lambda_1+\lambda_2) =$

$\lambda_1 + \lambda_2 - 2\lambda_2 + 2\lambda_1 - \lambda_2 =$

$= 3\lambda_1 - 2\lambda_2 = 0$

$\Rightarrow \lambda_2 = \frac{3}{2}\lambda_1$

$\left(\frac{5}{2}\lambda_1, -\frac{3}{2}\lambda_1, -\frac{1}{2}\lambda_1, 0\right)$

$X \cap V = \langle (5,-3,-1,0) \rangle$

$\dim(X \cap V) = 1$

b) $(5,-3,-1,0) = a(0,1,0,1) + b(0,0,1,1) + c(1,0,0,0)$

$\begin{cases} c = 5 \\ a = -3 \\ b = -1 \\ a + b = 0 \end{cases}$ NO

c) Base e dim $W+V = \mathbb{R}^4$!!

$V = \langle (-1,1,1,1), (1,0,1,0), (0,1,2,1) \rangle$

$\langle (0,1,0,1), (0,0,1,1), (1,0,0,0), (-1,1,1,1) \rangle$ ~~$(1,0,1,0), (0,1,2,1)$~~

$\dim(W+V) = 4$

$$\rightarrow \begin{pmatrix} 0&0&1&-1&1&0 \\ 1&0&0&1&0&1 \\ 0&1&0&1&1&2 \\ 1&1&0&1&0&1 \end{pmatrix} \rightarrow \begin{pmatrix} 1&0&0&1&0&1 \\ 0&1&0&0&0&0 \\ 0&0&0&1&1&2 \\ 0&0&1&-1&1&0 \end{pmatrix} \rightarrow \begin{pmatrix} 1&0&0&1&0&1 \\ 0&1&0&0&0&0 \\ 0&0&1&-1&1&0 \\ 0&0&0&1&1&2 \end{pmatrix}$$

$\text{rk} = 4$

Es. 4

$$M = \begin{pmatrix} 1&-2&-1&0 \\ 1&2&2&1 \\ 2&0&1&1 \end{pmatrix}$$

Sia $f: \mathbb{R}^4 \to \mathbb{R}^3$ appl. associata rispetto alle basi canoniche

a) $\text{rk}\, M = 2$

b) dim e base $\text{Im} f$

$\text{rk} M = \dim \text{Im} f = 2$ $\quad \langle (1,1,2), (-1,1,0) \rangle$

c) dim e base $\text{Ker} f$

$\dim(\text{Ker} f) = \dim(\mathbb{R}^4) - \dim(\text{Im} f) = 2$

base $\rightarrow$ $Mx = 0$

$$\begin{pmatrix} 1&-2&-1&0 \\ 1&2&2&1 \\ 2&0&1&1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = 0$$

z e t sono le variabili libere

$x = 2y + z$

$\begin{cases} x - 2y - z = 0 \\ x + 2y + 2z + t = 0 \end{cases}$ $\rightarrow$ $2y$

$4y + 3z + t = 0$

$y = -\frac{3}{4}z - \frac{1}{4}t$

$x = 2y + z = -\frac{1}{2}z - \frac{1}{2}t$

$= -\frac{3}{2}z - \frac{1}{2}t + z =$

$\left(-\frac{1}{2}z - \frac{1}{2}t, -\frac{3}{4}z - \frac{1}{4}t, z, t\right) =$

$\left(-\frac{2}{2}z - \frac{2}{2}t, -\frac{3}{2}z - t, \frac{4}{4}z, \frac{4}{4}t\right) = z(-2,-3,4,0) + t(-2,-1,0,4).$

$\text{Ker} f = \langle (-2,-3,4,0), (-2,-1,0,4) \rangle$

Es. 5

$$A = \begin{pmatrix} 0&1&0 \\ 0&0&1 \\ 1&0&0 \end{pmatrix}$$

$$p(\lambda) = \det \begin{pmatrix} -\lambda&1&0 \\ 0&-\lambda&1 \\ 1&0&-\lambda \end{pmatrix} = -\lambda \cdot (\lambda^2) - 1(-1) =$$

$= \lambda^3 + 1 =$

$= (1-\lambda^3)$

$\lambda = 1$

$m_a = 1$ → $\sum m_a < 3$!!

NON DIAG. in $\mathbb{R}$

m_a in $\mathbb{C}$?

b) $\lambda^3 = 1$

$\lambda = 1$

$|\lambda|^1 (\cos(\theta) + i\sin(\theta))$

$-\frac{1}{2} + i\frac{\sqrt{3}}{2}$

$\theta = ?$

$e^{i\cdot 0} = 1$

$e^{i\frac{2}{3}\pi} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$

$e^{i\frac{4\pi}{3}} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

$\sqrt[3]{1} = 1$

$\frac{2\pi}{3}$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2}+i\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & -\frac{1}{2}-i\frac{\sqrt{3}}{2} \end{pmatrix}$$